

# **Lesson 8 RRCLite PWM Servo Motion Speed Control Principle and Source Code Analysis**

## **1. PWM Servo Description**

A servo is a common motor drive device used to control the position, speed, and acceleration in a mechanical system. It typically consists of a motor, a control circuit, and a feedback system.

In terms of type, servos can be categorized into analog servos, PWM (pulse width modulation) servos, and bus servos. This section mainly focuses on PWM servos.

PWM servos use pulse width modulation signals to control the angular position of the servo.

In simple terms, a PWM servo controls the angular position by varying the pulse width of the signal. The range of the pulse width determines the extreme angular positions of the servo, while the period of the control signal determines the refresh frequency of the signal. By adjusting the duration of the pulse width, precise control of the servo position can be achieved.

PWM achieves a level state between 0 and 1, meaning that when PWM is output, the voltage can be any value between 0 and 3.3V/5V.

When using PWM control, it is essential to know two parameters of the object being controlled:

Its PWM frequency (corresponding to the timer cycle)

Its operating pulse width (corresponding to the PWM duty cycle)

The control signal enters the signal modulation chip from the receiver channel to obtain a DC bias voltage. Inside, a reference circuit generates a reference signal with a period of 20ms and a width of 1.5ms. The obtained DC bias voltage is compared with the voltage of the potentiometer to produce a voltage

difference output. Finally, the positive and negative outputs of the voltage difference are sent to the motor driver chip to determine the motor's forward and reverse rotation. When the motor speed is constant, the potentiometer is driven to rotate through the cascade reduction gear, ensuring the voltage difference becomes zero, and the motor stops rotating.

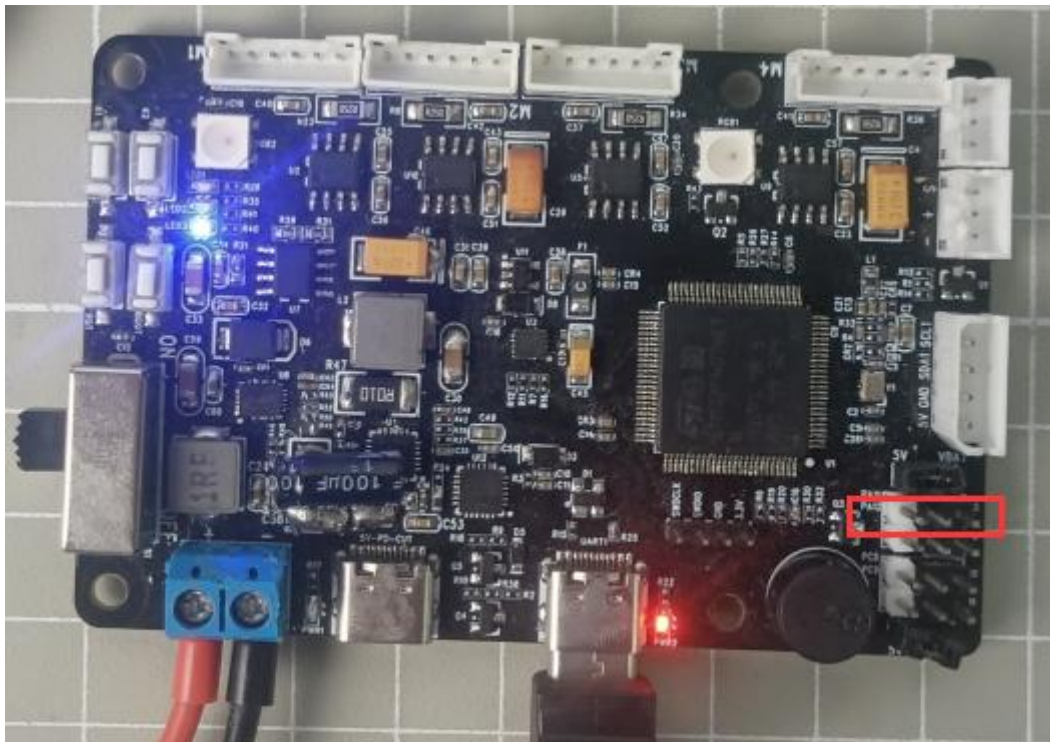
## 2. Wiring Method:

### 2.1 Required Stuffs:

1. stm32 controller
2. LD-1501MG digital servo

### 2.2 Connection Instructions

1. The diagram below shows the reserved interface J3 provided by the STM32 main control board for the PWM servo.



The reserved interface for PWM servos is J3, as shown below. There are two power supply options for the servo: VBAT (battery voltage) and 5V (+), as indicated by the green box in the diagram.



2. Connect the servo's 3-pin cable to the corresponding J2 interface, as shown below:

The connection method is as follows:

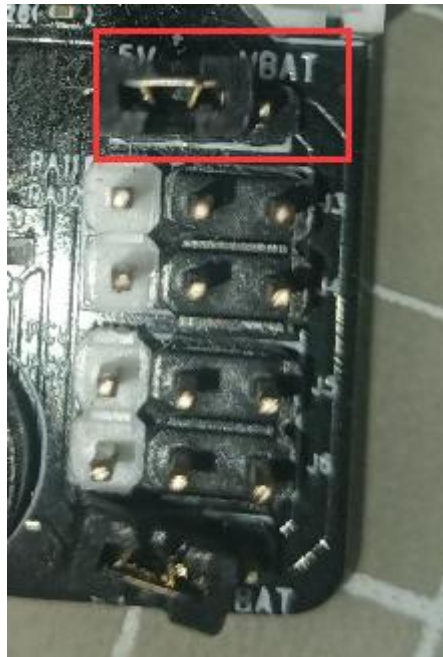
White -> Signal line (SUBS)

Red -> Power supply (+5V)

Black -> Ground (-)



3. Select the 5V power supply for J2 and connect the 5V pin with the middle pin on J2 using a jumper cap, as shown in the diagram below.



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Attention: The servo motor must be connected according to the positions shown in the document diagram. Avoid connecting it incorrectly or mistakenly to prevent burning out the motor. Our company will not take any responsibility!!!

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### **3. Program Download**

The sample program is 'RosRobotControllerLite\_ros'.

After downloading the program (refer to '2. STM32 Development Fundamentals/Lesson 3 Project Compilation and Download' for details on how to download the program), you need to press the RESET button to reset the microcontroller, which will then start executing the new program.

### **4. Program Outcome**

After pressing reset, the servo will first initialize to 90°, then move to 0°, and finally rotate to 180° at a speed of 60 degrees per second.

### **5. Code Analysis**

This program is only for speed control. For detailed PWM servo control

routines, please refer to “Lesson 7 RRCLite PWM Servo Control Principle and Source Code Analysis”.

```
void app_task_entry(void *argument)
{
    extern osTimerId_t led_timerHandle;
    extern osTimerId_t buzzer_timerHandle;
    extern osTimerId_t button_timerHandle;
    extern osTimerId_t battery_check_timerHandle;
    extern osMessageQueueId_t moving_ctrl_queueHandle;

    leds_init();
    motors_init();
    pwm_servos_init(); //return the servo to the neutral position with the default pulse width 1500 and rotation angle 90°
    serial_servo_init();
    buzzers_init();
    buttons_init();
    WS2812b_Configuration();

    button_register_callback(buttons[0], button_event_callback);
    button_register_callback(buttons[1], button_event_callback);

    osTimerStart(led_timerHandle, LED_TASK_PERIOD);
    osTimerStart(buzzer_timerHandle, BUZZER_TASK_PERIOD);
    osTimerStart(button_timerHandle, BUTTON_TASK_PERIOD);
    osTimerStart(battery_check_timerHandle, BATTERY_TASK_PERIOD);
    packet_handle_init();

    uint32_t time = 3000; // motion time (ms)
```

The program is located in the app.c file under /Hiwonder/System.

The pwm\_servo\_set\_position() function controls the rotation of the servo. The required parameters are defined as follows:

Parameter 1: Servo object

Parameter 2: Target angle of the servo

Parameter 3: Movement time

The rotation speed can be obtained by dividing the rotation angle by the movement time. The program logic is as follows:

- ① Initialize the servo;
- ② Set the servo to a known initial angle;
- ③ Delay to allow the servo to complete the movement. The delay time must be greater than or equal to the movement time of the servo, otherwise, the servo cannot reach the target position.
- ④ Move the servo to the target angle and specify the movement time.
- ⑤ Delay to allow the servo to complete the movement.

The servo movement speed can be calculated by dividing the known angle difference by the movement time.